

Week 1 & 2 – GA 1

Common Data for Questions (1) and (2)

Let $X \sim \text{Geometric}(p)$. Recall that the discrete random variable X has the following probability mass function:

$$P(X = x) = (1 - p)^{x-1} p$$

1) Using Markov's inequality, obtain an upper bound for $P\left(X \geq \frac{2}{p}\right)$.

0.5

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 0.5

1 point

2) Recall that the obtained answer is a loose upper bound. For the instance of $p = 0.2$, evaluate the true probability of $P\left(X \geq \frac{2}{p}\right)$. What is the absolute difference between this probability and the Markov upper bound? Enter your answer correct to three decimal places.

0.366

$$1. \quad P\left(X \geq \frac{2}{p}\right) = ?$$

$$\text{Markov inequality} \Rightarrow P(X \geq a) \leq \frac{E[X]}{a}$$

$$\text{For geometric R.V.s} \Rightarrow \mu = \frac{1}{p}$$

$$P\left(X \geq \frac{2}{p}\right) \Rightarrow a = \frac{2}{p} \\ \leq \frac{1 \times 1}{2 \times 2} \leq 0.5$$

$$2. \quad p = 0.2 \Rightarrow P\left(X \geq \frac{2}{0.2}\right) \Rightarrow P(X \geq 10)$$

$$P(X = x) = (1 - p)^{x-1} \times p \\ = (0.8)^9 \times 0.2 \approx 0.134$$

$$\text{Absolute diff.} \Rightarrow |0.5 - 0.134| = 0.366$$

3.

$$\text{Chebyshev} \Rightarrow P(|X - \mu| \geq a) \leq \frac{\sigma^2}{a^2}$$

$$P(|X - \mu| \geq a) \geq 1 - \frac{\sigma^2}{a^2}$$

Questions (3) to (5) form a group

3) Let $X \sim \text{Normal}(0, 1)$. Using Chebyshev's inequality, we get the following bound:

$$P(-2 \leq X \leq 2) \geq p$$

Enter the value of p . Enter -1 if p cannot be obtained.

0.75

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 0.75

$$\sigma^2 = 1$$

$$\mu = 0$$

$$P(X \geq 2) \geq p$$

$$p = 1 - \frac{1}{4} = \frac{3}{4} = 0.75$$

$$4. E[X] = 0$$

4) Let $X \sim \text{Normal}(0, 1)$. Using Markov's inequality, we get the following bound:

$$P(X^2 \geq 2) \leq p$$

Enter the value of p . Enter -1 if p cannot be obtained.

0.5

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 0.5

$$P(Y \geq a) \leq \frac{E[Y]}{a}$$

$$Y = X^2 \Rightarrow P(X^2 \geq 2) \leq p \leq \frac{E[X^2]}{2}$$

$$\text{Var}(X) = E[X^2] + (E[X])^2$$

$$E[X^2] = 1 - 0 = 1$$

$$P(X^2 \geq 2) \leq \frac{1}{2} \leq 0.5$$

$$5. E[X] = 0$$

$$E[X^2] = 1$$

5) Let $X \sim \text{Normal}(0, 1)$. The 4th Moment of X i.e. $E[X^4] = 3$. Using Chebyshev's inequality, we get the following bound:

$$P(X^2 \geq 2) \leq q$$

Enter the value of q . Enter -1 if q cannot be obtained.

2

$$Y = X^2$$

$$E[Y] = E[X^2] = 1 \text{ (prev.)}$$

$$E[Y^2] = E[X^4] = 3 \text{ (given)}$$

$$\begin{aligned} \text{Var}(Y) &= E[Y^2] - (E[Y])^2 \\ &= 3 - 1 = 2 \end{aligned}$$

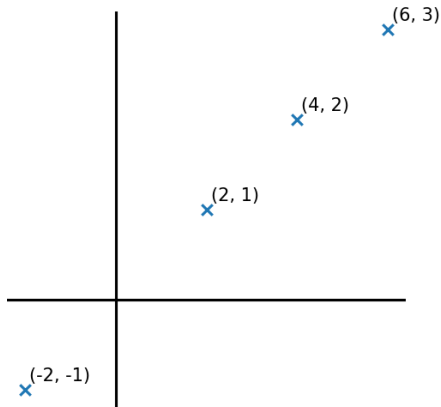
$$P(|Y - E[Y]| \geq a) \leq \frac{\sigma^2}{a^2}$$

$$Y \geq 2 \Rightarrow Y - 1 \geq 1 \Rightarrow |Y - 1| \geq 1$$

$$P(Y \geq 2) \leq P(|Y - 1| \geq 1) \leq \frac{\sigma^2}{a^2} = \frac{2}{1} = 2 = q$$

6.

6) Consider the following dataset:

Does there exist a distance preserving (isometric) low rank linear transformation f for the given set of points?
☒ Yes
☐ No

$$\begin{matrix} x & y \\ -2 & -1 \\ 2 & 1 \\ 4 & 2 \\ 6 & 3 \end{matrix}$$

$$y = \frac{x_{\text{point}}}{2}$$

So that means all of these 4 points lie on the same straight line through the origin.

$f(x, y) = \frac{2x + y}{\sqrt{5}}$ Yes, they lie 1-D subspace of $\mathbb{R}^2$ so, they are isometric.

7.

7) Let $\mathbf{a}, \mathbf{b} \in \mathbb{R}^{10}$ be two random vectors whose entries are i.i.d standard normal variables. Let $\mathbf{c} = \mathbf{a} - \mathbf{b}$. Let t be the sum of all eigenvalues of the matrix $\mathbf{c}\mathbf{c}^T$. Observe that t is a random variable. What is $\mathbb{E}[t]$?

20

for a matrix M ,sum of eigenvalues of M = trace of M

$$t = \sum_{i=1}^{10} \lambda_i(M) = \text{trace of } M = \text{trace of } \mathbf{c}\mathbf{c}^T = \mathbf{c}^T \mathbf{c} = \|\mathbf{c}\|^2$$

$$t = \|\mathbf{c}\|^2 = \sum_{i=1}^{10} c_i^2$$

$$c_i = a_i - b_i \quad a, b \sim N(0, 1) \text{ i.i.d}$$

$$c_i \sim N(0, 2)$$

$$t = \sum_{i=1}^{10} c_i^2$$

$$Z \sim N(0, \sigma^2)$$

$$\mathbb{E}[Z^2] = \text{Var}(Z) = \sigma^2 = 2$$

$$\mathbb{E}[c_i^2] = 2$$

$$\begin{aligned} \mathbb{E}[t] &= \mathbb{E}\left[\sum_{i=1}^{10} c_i^2\right] = \sum_{i=1}^{10} \mathbb{E}[c_i^2] = \\ &= 10 \times 2 = 20 \end{aligned}$$

no. of pts in class $i \rightarrow n_i$ ($i = 1 \dots 10$)

$$\sum_{i=1}^{10} n_i = 1000$$

$$n_1 + n_3 + n_5 = 200$$

$$n_2 + n_4 + n_3 = 400$$

$$p = \frac{n_1 + n_2 + n_3 + n_4 + n_5}{1000}$$

Common Data for Questions (8) to (10)

Consider a training dataset for a multi-class classification problem that has 1000 data points with 10 classes, where each class contributes at least one data point. Classes 1, 3 and 5 together contribute 200 data points. Classes 2, 3, 4 together contribute 400 data points. A data point is picked at random. p is the probability that it belongs to any one of the following classes: 1, 2, 3, 4 or 5.

8) Which of the following is true?

☐ $p \leq 0.5$

☐ $p \leq 0.2$

☒ $p \leq 0.6$

☐ $p \leq 0.4$

Yes, the answer is correct.

Score: 1

Accepted Answers:

$p \leq 0.6$

$$8. \quad 200 + 400 - n_3 = 600 - n_3$$

$$n_1 + n_2 + n_3 + n_4 + n_5 \leq 600$$

$$p \leq \frac{600}{1000} \leq 0.6$$

9) Across all possible datasets that satisfy the conditions given in the common data, what is the largest possible value that p can assume? Your answer should be exact, with three places after the decimal point.

0.599

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 0.599

10) Across all possible datasets that satisfy the conditions given in the common data, what is the smallest possible value that p can assume? Your answer should be exact, with three places after the decimal point.

0.402

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 0.402

$$9. \text{ we saw that } \Rightarrow p = \frac{600 - n_3}{1000}$$

$$\max p \Rightarrow \min n_3 \Rightarrow \text{but } n_i \geq 1$$

$$\therefore, n_3 = 1$$

$$p = \frac{600 - 1}{1000} = 0.599.$$

$$10. \min p \Rightarrow \max n_3$$

$$\textcircled{1} \quad n_1 = 1, n_5 = 1, n_3 = 198$$

$$n_2 \geq 1, n_4 \geq 1, n_3 \leq 398$$

$$\textcircled{2} \quad n_1 = 1, n_5 = 1, n_2 = 1, n_4 = 201$$

$$n_3 = 198$$

$$p_{\min} = \frac{600 - 198}{1000} = 0.402$$

$$11. X_i = \begin{cases} 1 & \text{heads} \\ 0 & \text{tails} \end{cases} \quad E[X_i] = p \quad \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

Common Data for Questions (11) to (12)

A biased coin, whose probability of landing heads equals p , is tossed 10 times. Let q be the probability that the fraction of heads observed is at least $0.1 + p$.

11) Using Hoeffding's inequality, we get $q \leq r$. Enter the value of r correct to two places after the decimal.

0.82

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Range) 0.81,0.83

$$E[\bar{X}] = p$$

$$P(\bar{X} - E[\bar{X}] \geq \epsilon) \leq e^{-2n\epsilon^2}$$

$$P(\bar{X} \geq p + 0.1) = P(\bar{X} - p \geq 0.1)$$

$$\leq e^{-2(10)(0.1)^2}$$

$$\approx 0.82$$

1 point

12.

12) Consider the same setup with two changes. The coin is now tossed n times. q' is the probability that the fraction of heads observed is at least $0.01 + p$. Let the bound obtained from Hoeffding be $q' \leq r'$. What is the smallest value of n that would guarantee that $r' \leq e^{-0.2}$? Your answer should be an integer.

1000

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 1000

$$q' = P(\bar{X} - p \geq 0.01)$$

$$q' \leq r' = e^{-2n \times 0.0001}$$

$$= e^{-0.0002n}$$

$$e^{-0.0002n} \leq e^{-0.2}$$

$$n \geq \frac{0.2}{0.0002} = 1000$$

1 point

13.

Common Data for Questions (13) to (14)

Consider an outlier detection algorithm that is deployed by a data-science company. Assume that the number of outliers it detects on a given day is a random variable with mean 50.

$$\mu = 50$$

$$\text{Var} = 25 = \sigma^2$$

13) Let p be the probability that the number of outliers exceeds 75. Choose the tightest bound with the information available at your disposal.

1 point

- ☐ $p \geq \frac{1}{3}$
- ☐ $p \leq \frac{1}{3}$
- ☐ $p \geq \frac{2}{3}$
- ☒ $p \leq \frac{2}{3}$

Yes, the answer is correct.

Score: 1

Accepted Answers:

$p \leq \frac{2}{3}$

$$p = P(X > 75) \leq \frac{E[X]}{a} \leq \frac{50}{75} \leq \frac{2}{3}$$

$$p \leq \frac{2}{3}$$

14.

14) The variance of the number of outliers is equal to 25. Let q be the probability that the number of outliers lies between 40 and 60, end-points excluded. Choose the tightest bound with the information available at your disposal.

1 point

- ☐ $q \leq \frac{3}{4}$
- ☒ $q \geq \frac{3}{4}$
- ☐ $q \leq \frac{1}{4}$
- ☐ $q \geq \frac{1}{4}$

Yes, the answer is correct.

$$P(|X - \mu| \geq a) \leq \frac{\sigma^2}{a^2}$$

$$\text{or } P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

$$q = P(40 < X < 60)$$

$$= P(|X - 50| < 10)$$

$$= P(|X - 50| \geq 10) \leq \frac{1}{2^2} = \frac{1}{4}$$

$$q = P(|X - 50| < 10) \geq 1 - \frac{1}{4} = \frac{3}{4}$$

$$q \geq \frac{3}{4}$$

15.

$$R = 100 \times 10000 \quad M = \frac{R}{10}, \mu \in \mathbb{R}^{10000}$$

$$v = Mu$$

Questions (15) and (16) form a group

15) Let R be a 100×10000 random matrix whose entries are i.i.d standard normal random variables. Let $M = \frac{R}{10}$. $u \in \mathbb{R}^{10000}$ is a non-zero, fixed vector and $v = Mu$. Which of the following is true? 1 point

- ☐ $\|v\|^2 = \|u\|^2$
- ☐ $E[\|u\|^2] = \|v\|^2$
- ☒ $E[\|v\|^2] = \|u\|^2$
- ☐ $E[\|v\|^2] = \|v\|^2$

Yes, the answer is correct.
Score: 1
Accepted Answers:
 $E[\|v\|^2] = \|u\|^2$

$$v = Mu = \frac{R}{10}u \quad v_i = \frac{1}{10} \sum_j R_{ij} u_j \quad E[x_i^T u] = 0$$

$$Var(\cdot) = \|u\|^2$$

$$E[\|v\|^2] = \sum_{i=1}^{100} E[v_i^2] \Rightarrow E[v_i^2] = \frac{\|u\|^2}{100}$$

16.

16) If $x = \frac{10 \cdot v}{\|u\|}$, then which of the following is true?

- ☐ $\|x\|^2 = 100$
- ☐ $E[\|x\|^2] = 100\|v\|^2$
- ☒ $E[\|x\|^2] = 100$
- ☐ $E[\|x\|^2] = \|x\|^2$

$$E[\|v\|^2] = \frac{100 \times \|u\|^2}{100} = \|u\|^2$$

$$x = \frac{10 \cdot v}{\|u\|} \Rightarrow \|x\|^2 = \frac{100 \|v\|^2}{\|u\|^2}$$

$$E[\|x\|^2] = \frac{100}{\|u\|^2} E[\|v\|^2] = \frac{100}{\|u\|^2} \|u\|^2 = 100$$

17.

17) Consider a dataset D of n data-points in $\mathbb{R}^d$. Given $0 < \epsilon < 1$, let f be a linear map from $\mathbb{R}^d \rightarrow \mathbb{R}^k$, an approximate isometry as guaranteed by the JL lemma. Which of the following are true? 1 point

- ☒ $f(0) = 0$
- ☒ $f(x+y) = f(x) + f(y)$, for all $x, y \in \mathbb{R}^d$
- ☒ $(1-\epsilon)\|x-y\|^2 \leq \|f(x) - f(y)\|^2 \leq (1+\epsilon)\|x-y\|^2$, for all $x, y \in D$
- ☐ $(1-\epsilon)\|x-y\|^2 \leq \|f(x) - f(y)\|^2 \leq (1+\epsilon)\|x-y\|^2$, for all $x, y \in \mathbb{R}^d$

Yes, the answer is correct.
Score: 1
Accepted Answers:
 $f(0) = 0$
 $f(x+y) = f(x) + f(y)$, for all $x, y \in \mathbb{R}^d$
 $(1-\epsilon)\|x-y\|^2 \leq \|f(x) - f(y)\|^2 \leq (1+\epsilon)\|x-y\|^2$, for all $x, y \in D$

since f is linear so $\forall x \quad f(0) = f(0 \cdot x) = 0$
again it is a linear map,
these $\begin{cases} f(x+y) = f(x) + f(y) \\ f(\alpha x) = \alpha f(x) \end{cases} \forall x, y \in \mathbb{R}^d$
will hold true.

3rd option is JL lemma it stands true for all x, y pairs in dataset D .
This is surely not about $\mathbb{R}^d$.

$$D \subset \mathbb{R}^d$$

$$D^+ : \|x\|^2 \leq \kappa_1^2$$

$$D^- : \|x\|^2 \geq \kappa_2^2$$

$$(1-\epsilon)\|x\|^2 \leq \|f(x)\|^2 \leq (1+\epsilon)\|x\|^2$$

18.

Common Data for Questions 18) to 20)

Consider a dataset D of n data-points for a binary classification problem in $\mathbb{R}^d$ that is non-linearly separable with a spherical decision boundary. To be more precise, the set D^+ of all positive data-points belongs to the region $\|x\|^2 \leq r_1^2$ and the set D^- of all negative data-points belongs to the region $\|x\|^2 \geq r_2^2$, with $r_2 > r_1$. To get a visual picture, think about $d=3$ and consider two concentric spheres.

It is also given that the origin is one of the data-points, namely $0 \in D$. Using the JL lemma, let f be a linear map that transforms all data-points to $\mathbb{R}^k$ for a suitable choice of ϵ and k .

18) With the available information, which of the following statements is true concerning the positive data-points in $\mathbb{R}^k$ after the transformation? The norms are with respect to $\mathbb{R}^k$.

1 point

- ☐ $\|f(x)\|^2 \leq r_1^2$, for all $x \in D^+$
- ☐ $\|f(x)\|^2 \leq (1-\epsilon)r_1^2$, for all $x \in D^+$
- ☐ $\|f(x)\|^2 \geq (1+\epsilon)r_2^2$, for all $x \in D^+$
- ☒ $\|f(x)\|^2 \leq (1+\epsilon)r_1^2$, for all $x \in D^+$

Yes, the answer is correct.

Score: 1

Accepted Answers:

$\|f(x)\|^2 \leq (1+\epsilon)r_1^2$, for all $x \in D^+$

$$\|f(x)\|^2 \leq (1+\epsilon)\|x\|^2$$

$$\leq (1+\epsilon)\kappa_1^2$$

19.

19) With the available information, which of the following statements is true concerning the negative data-points in $\mathbb{R}^k$ after the transformation? The norms are with respect to $\mathbb{R}^k$.

1 point

- ☐ $\|f(x)\|^2 \geq r_2^2$, for all $x \in D^-$
- ☐ $\|f(x)\|^2 \leq (1-\epsilon)r_2^2$, for all $x \in D^-$
- ☒ $\|f(x)\|^2 \geq (1-\epsilon)r_2^2$, for all $x \in D^-$
- ☐ $\|f(x)\|^2 \leq (1+\epsilon)r_1^2$, for all $x \in D^-$

Yes, the answer is correct.

Score: 1

Accepted Answers:

$\|f(x)\|^2 \geq (1-\epsilon)r_2^2$, for all $x \in D^-$

$$(1-\epsilon)\|x\|^2 \leq \|f(x)\|^2$$

$$(1-\epsilon)\kappa_2^2 \leq \|f(x)\|^2$$

20.

20) If $\frac{r_1}{r_2} = 0.2$, find an upper bound for ϵ that ensures that the dataset continues to be spherically separable in the lower dimensional space $\mathbb{R}^k$. Enter your answer correct to two places after the decimal point.

0.92

Yes, the answer is correct.

Score: 1

$$\frac{\kappa_1}{\kappa_2} = 0.2$$

$$\kappa_1^2 = 0.04 \kappa_2^2$$

$$(1+\epsilon)0.04\cancel{\kappa_2^2} < (1-\epsilon)\cancel{\kappa_2^2}$$

$$0.04(1+\epsilon) < (1-\epsilon)$$

$$\epsilon < \frac{0.96}{1.04} = 0.92$$

$$\epsilon < 0.92$$